

诚信应考，考试作弊将带来严重后果！

华南理工大学本科生期末考试

2024-2025-2 学期章节试卷

《Calculus Chapter Review-Line integral and Surface integral》

注意事项：1. 开考前请将密封线内各项信息填写清楚；

2. 所有答案请直接答在试卷上；

3. 考试形式：闭卷；

4. 本试卷共 5 大题，满分 50 分，考试时间 50 分钟。

1, Let G be the sphere $x^2 + y^2 + z^2 = a^2$. Evaluate $\iint_G (x^2 + y^2 + z^2)^5 dS$

2, Let G be the sphere $x^2 + y^2 + z^2 = a^2$. Evaluate $\iint_G \frac{x + y^3 + \sin z}{1 + z^4} dS$

3, Evaluate $\int_{(0,0,0)}^{(1,1,1)} (6xy^3 + 2z^2) dx + 9x^2y^2 dy + (4xz + 1) dz$

4, Calculate $\int_{(2,1)}^{(6,3)} \frac{x^3}{(x^4 + y^4)^2} dx + \frac{y^3}{(x^4 + y^4)^2} dy$

5, Evaluate line integral $\oint_C (e^{3x} + 2y) dx + (x^2 + \sin y) dy$, where C is the positive rectangle with vertices $(2,1), (6,1), (6,4), (2,4)$

6, Evaluate line integral $\oint_C (x^2 + 4xy) dx + (2x^2 + 3y) dy$, where C is the positive closed curve formed by $9x^2 + 16y^2 = 144$.

7, Let S be the solid bounded by $x^2 + z^2 \leq y^2$, $x^2 + y^2 + z^2 \leq 1$, $y \geq 0$ and let $\vec{n}$ be the outer unit normal to the boundary ∂S . Find the surface integral

$$\iint_{\partial S} \vec{F} \cdot \vec{n} dS = \iint_{\partial S} (x^3 + y) dy dz + (y^3 + z) dz dx + (x + z^3) dx dy$$

8, Let C be the positive closed curve formed by $(x-a)^2 + (y-a)^2 = 1$, and $f(x)$ is a positive continuous function. Prove that the line integral

$$\oint_C \frac{x}{f(y)} dy - yf(x) dx \geq 2\pi.$$